

Maximum number of linear regions induced by n ReLU neurons

Basically this is a combinatorial problem where we have to find the maximum amount of regions when the lines are in *general position*, that is,

- no two lines are parallel;
- no three lines intersect at the same point.

If there are no lines, the plane consists of a single region.

$$R(0) = 1.$$

When the first line is added, the plane is divided into two regions.

$$R(1) = 2.$$

Suppose we already have $n - 1$ lines in general position and we add line number n . Since the new line intersects each of the $n - 1$ previous lines exactly once, $n - 1$ intersection points appear on it. These points divide the line into n segments.

For example,

Intersections	Segments
0	1
1	2
2	3
3	4

In general, a line with $n - 1$ intersection points is divided into n segments.

Each of these segments passes through exactly one region of the plane. As it passes through that region, the segment splits it into two. Therefore,

each segment creates exactly one new region. Since there are n segments, we obtain the recurrence relation

$$R(n) = R(n-1) + n.$$

Expanding the recurrence,

$$\begin{aligned} R(n) &= R(n-1) + n \\ &= R(n-2) + (n-1) + n \\ &= R(n-3) + (n-2) + (n-1) + n \\ &\vdots \\ &= R(0) + 1 + 2 + \cdots + n. \end{aligned}$$

Since

$$R(0) = 1,$$

we obtain

$$R(n) = 1 + \sum_{k=1}^n k.$$

Finally, using the well-known identity

$$\sum_{k=1}^n k = \frac{n(n+1)}{2},$$

we conclude that

$$\boxed{R(n) = \frac{n(n+1)}{2} + 1.}$$

The first values are

n	0	1	2	3	4	5
$R(n)$	1	2	4	7	11	16

We notice that the number of hypotheses $R(n)$ grows quadratically in relation to the number of neurons n .